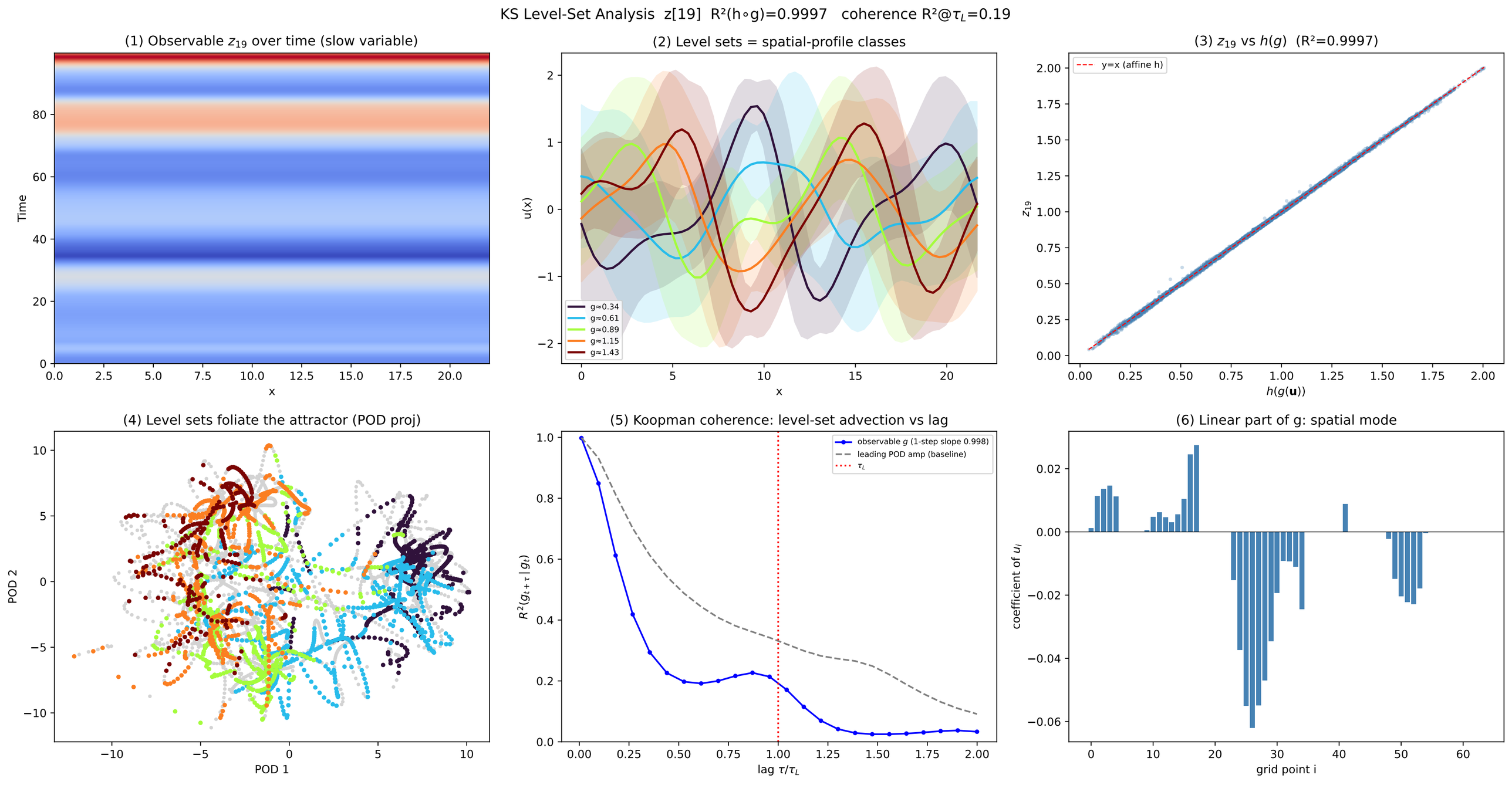

KS Level-Set Analysis z[19] R²(h∘g)=0.9997 coherence R²@τ_L=0.19
(1) Observable z_{19} over time (slow variable)
Time
x
(2) Level sets = spatial-profile classes
u(x)
x
g≈0.34
g≈0.61
g≈0.89
g≈1.15
g≈1.43
(3) z_{19} vs $h(g)$ (R²=0.9997)
y=x (affine h)
z_{19}
$h(g(\mathbf{u}))$
(4) Level sets foliate the attractor (POD proj)
POD 2
POD 1
(5) Koopman coherence: level-set advection vs lag
observable g (1-step slope 0.998)
leading POD amp (baseline)
τ_L
$R^2(g_{t+\tau}\,|\,g_t)$
lag τ/τ_L
(6) Linear part of g: spatial mode
coefficient of u_i
grid point i